

# Analysis group meeting

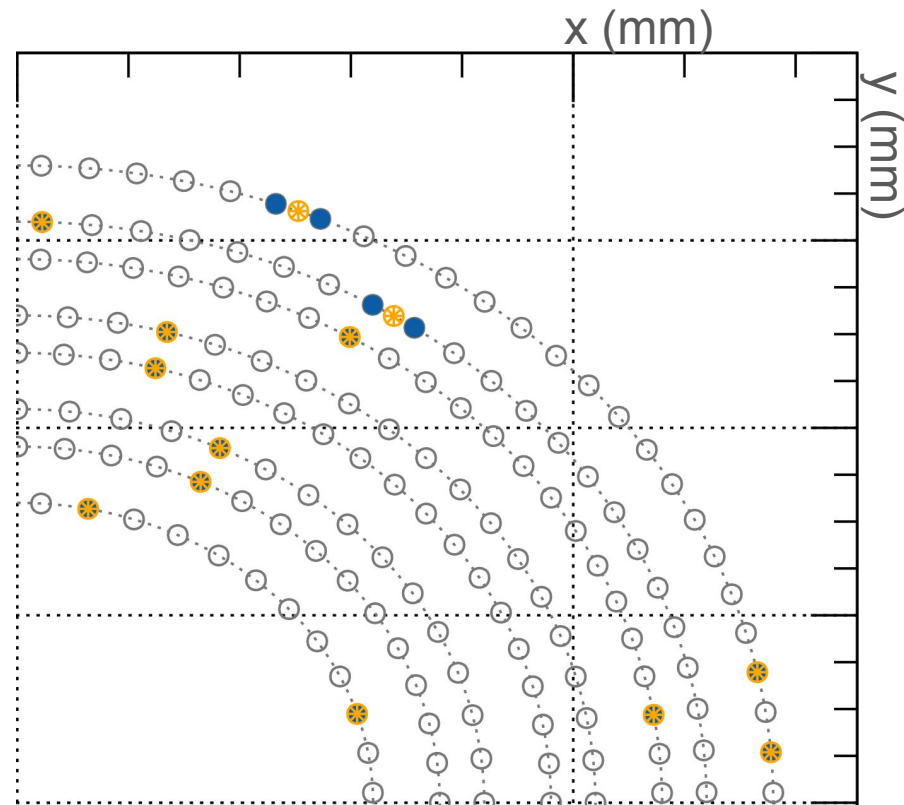
Mathieu Ouillon (Mississippi State University)

November 26<sup>th</sup>, 2024

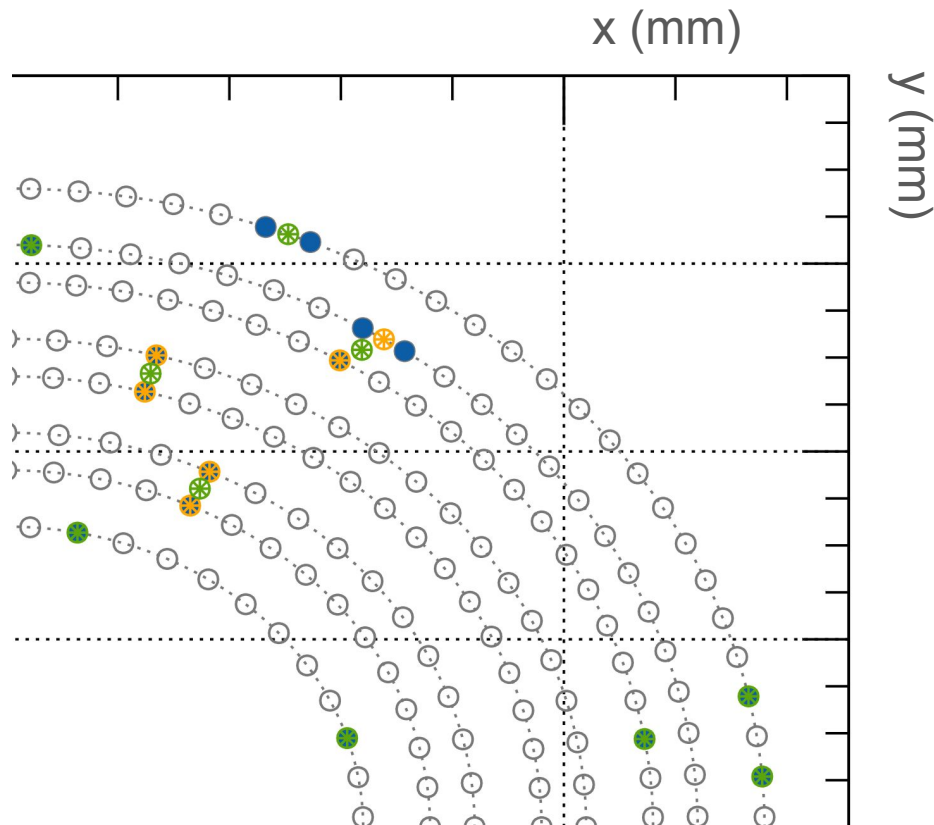


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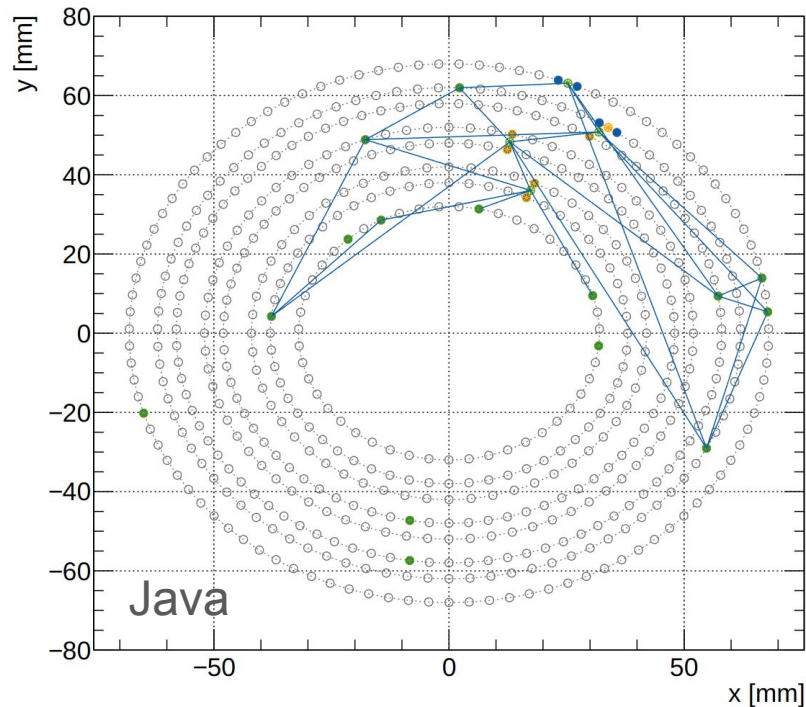
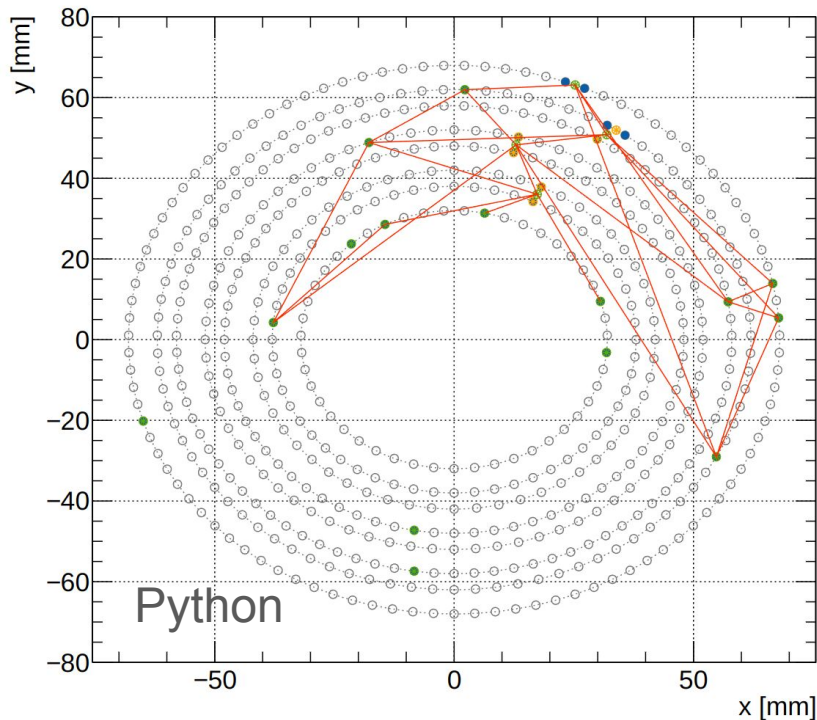
- Implementation of the clustering in COATJAVA:
  - Raw hits in blue.
  - Merge hits that on the same layer and 1 wire away together into **precluster** (orange circle).
  - Merge precluster that are on the same superlayer and 8mm away into **super-precluster** (green circle).



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- Find all possible tracks: implementation in COATJAVA



- Implementation of the AI in COATJAVA, use Deep Java Library

Prediction of python AI:

4.1537e-06, 1.7346e-06,  
4.2557e-07, 2.9792e-08,  
1.8523e-10, 9.8938e-01, ...

Prediction of AI in COATJAVA:

4.153695E-6, 1.7346407E-6,  
4.255732E-7, 2.9791678E-8,  
1.8522876E-10, 0.9893768, ...

- ALERT@22GeV
  - Compute statistical error for 50 PAC at  $2 \cdot 10^{35} \text{cm}^2 \text{s}^{-1}$
  - Change binning for the asymmetry
  - Incoherent DVCS on  $\text{He}^4$ :
    - TOPEG doesn't have a model for incoherent DVCS on helium4 (neither proton nor neutron)
- AI for ALERT:
  - Test the implement of the AI in COATJAVA in multi threaded (CLARA) (Doesn't expect any issues)
  - Recreate the efficiency/purity plot using COATJAVA